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Research Interests

- 📖 p -adic Galois representations
- 📖 p -adic transcendental number theory
- 📖 Formalization of arithmetic geometry

Employment

2025–2027 📖 **Postdoctoral Fellow, Institute for Theoretical Sciences, Westlake University**
Advisor: Huayi Chen

Education

- 2022–2025 📖 **Ph.D., Yau Mathematical Sciences Center, Tsinghua University**
Pure Mathematics
Thesis title: *The Field of Norms and $(\varphi, \Gamma_{\mathfrak{F}, K})$ -Modules for p -Adic False Tate Curve Extensions.*
Advisor: Lei Fu / Haoran Wang
- 2018 – 2021 📖 **M.Sc., Shanghai Center for Mathematical Sciences, Fudan University**
Pure Mathematics
Thesis title: *Uniformizer of the False Tate Curve Extension of $\mathbb{Q}_p$.*
Advisor: Shanwen Wang / Quanshui Wu
- 2014 – 2018 📖 **B.S., School of Mathematical Sciences, East China Normal University**
Mathematics and Applied Mathematics

Research Publications

Journal Articles

- 1 S. Wang and Y. Yuan, “Uniformizer of the false Tate curve extension of $\mathbb{Q}_p$ (II),” *International Journal of Number Theory*, vol. 20, no. 01, pp. 1–18, 2024. 🔗 DOI: 10.1142/S1793042124500015
- 2 S. Wang and Y. Yuan, “Uniformizer of the false Tate curve extension of $\mathbb{Q}_p$,” *The Ramanujan Journal*, vol. 58, no. 2, pp. 549–595, 2022. 🔗 DOI: 10.1007/s11139-021-00473-0

Preprints (all submitted)

- 1 Y. Yuan, *Formalization of non-archimedean functional analysis 1: Spherically complete spaces*, 2026. arXiv: 2601.21734 [math.NT].
- 2 Y. Yuan, *Descent of (φ, τ) -modules in characteristic p* , 2025. arXiv: 2511.17029 [math.NT].

- 3 S. Wang and Y. Yuan, *On the p -adic transcendence of $\sum_{k=1}^{\infty} p^{-1/p^k}$* , 2025. arXiv: 2509.24609 [math.NT].
- 4 Y. Yuan, *Formalization of Harder-Narasimhan theory*, 2025. arXiv: 2509.19632 [math.AG].
- 5 Y. Yuan, *Multivariable period rings of p -adic false Tate curve extension*, 2025. arXiv: 2505.24064 [math.NT].
- 6 S. Wang and Y. Yuan, *Hyper-algebraic invariants of p -adic algebraic numbers*, 2024. arXiv: 2402.15947 [math.NT].
- 7 S. Wang and Y. Yuan, *Truncated expansion of ζ_{p^n} in the p -adic Mal'cev-Neumann field*, 2023. arXiv: 2111.07127 [math.NT].

Talks

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| 2026, Apr. 9th |  Formalization of Non-Archimedean Functional Analysis
<i>School of Mathematical Sciences, East China Normal University</i> |
| 2026, Apr. 7th |  Formalization of Non-Archimedean Functional Analysis
<i>AI for Mathematics @ SIMIS</i>
<i>Shanghai Institute for Mathematics and Interdisciplinary Sciences</i> |
| 2025, Nov. 19th |  Formalization of Harder-Narasimhan Theorey
<i>Shanghai Institute for Mathematics and Interdisciplinary Sciences</i> |
| 2025, Jun. 13th |  Arithmetic of spherically complete valued fields in mixed characteristic
<i>Symposium on Number Theory and Algebraic Geometry of Spring 2025</i>
<i>Capital Normal University</i> |
| 2024, Nov. 1st |  Hyper-algebraic invariants of p-adic algebraic numbers
<i>Second Doctoral Conference on Frontiers of Mathematics</i>
<i>School of Mathematical Sciences, Peking University</i> |
| 2024, Oct. 16th |  Arithmetic of p-adic Mal'cev-Neumann field
<i>Number Theory Launch Seminar</i>
<i>Beijing Institute of Mathematical Sciences and Applications (BIMSA)</i> |
| 2023, Sep. 27th |  (φ, Γ)-modules for p-adic false Tate curve extension
<i>0-1 Seminar</i>
<i>Department of Mathematical Sciences, Tsinghua University</i> |

Awards

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| 2024 |  Outstanding Teaching Assistant in Undergraduate Fundamental Courses, Tsinghua University (Second Class) |
| 2023 |  Scholarship of Yau Mathematical Sciences Center (Second Class)
 Tsinghua University Comprehensive Excellence Scholarship (Second Class) |

Peer Review

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| 2024 |  Journal of Number Theory |
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Teaching Experience

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|--------------|---|
| 2025, Autumn |  TA for Foundation of Algebra and Analysis at Westlake University. |
| 2024, Autumn |  TA for an undergraduate course on Linear Algebra at Tsinghua University. |
| 2023, Autumn |  TA for an undergraduate course on Linear Algebra at Tsinghua University. |

Teaching Experience (continued)

2022, Autumn  TA for an undergraduate course on **Linear Algebra** at Tsinghua University.

Skills

Coding  Python (SageMath), $\text{\LaTeX}$, Lean4, Mathematica, ...